Hoeffding's Inequality: An Appendix-Grade Proof

1 Preliminaries: Hoeffding's lemma

Throughout, $c, C, C_1, C_2, \dots$ denote universal positive constants whose values may change from line to line and depend only on quantities specified in each statement's hypotheses. We write $\mathbb{P}[\cdot]$ for probability and $\mathbb{E}[\cdot]$ for expectation. For a real random variable Y, its *moment generating function* (MGF) is $\lambda \mapsto \mathbb{E}[e^{\lambda Y}]$, defined for all $\lambda \in \mathbb{R}$ whenever Y is bounded.

The single auxiliary ingredient is Hoeffding's lemma: a bounded, centered random variable has a sub-Gaussian MGF with the sharp variance proxy $(b-a)^2/4$. The leading constant 2 in the final tail bound is a direct consequence of the constant $1/8$ below.

Lemma 1.1 (Hoeffding's lemma)**.** *Let Y be a random variable with $\mathbb{E}[Y] = 0$ and $Y \in [a, b]$ almost surely, where $a < b$. Then for every $\lambda \in \mathbb{R}$,*

$$\mathbb{E}\Big[e^{\lambda Y}\Big] \;\le\; \exp\!\left(\frac{\lambda^2 (b-a)^2}{8}\right). \tag{1}$$

Proof. Fix $\lambda \in \mathbb{R}$ and let $\psi(\lambda) := \log \mathbb{E}[e^{\lambda Y}]$ be the cumulant generating function of Y; since Y is bounded, $\mathbb{E}[e^{\lambda Y}]$ is finite and ψ is infinitely differentiable on $\mathbb{R}$, with derivatives obtained by differentiating under the expectation. We compute the first two derivatives and bound the second.

Step 1 (boundary values).

$$\psi(0) \;=\; \log \mathbb{E}[e^0] \;=\; \log 1 \;=\; 0, \tag{2}$$

$$\psi'(\lambda) \;=\; \frac{\mathbb{E}[Y e^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]}, \qquad \text{so} \qquad \psi'(0) \;=\; \frac{\mathbb{E}[Y]}{\mathbb{E}[1]} \;=\; 0, \tag{3}$$

where the first line uses $\mathbb{E}[e^0] = 1$, and the second differentiates $\psi = \log \mathbb{E}[e^{\lambda Y}]$ and then sets $\lambda = 0$ using $\mathbb{E}[Y] = 0$.

Step 2 (the second derivative is a variance). Define, for each fixed λ, the tilted law Q_λ under which Y has density proportional to $e^{\lambda Y}$ with respect to its original law. Then

$$\begin{aligned}\psi''(\lambda) \;&=\; \frac{\mathbb{E}[Y^2 e^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]} - \left(\frac{\mathbb{E}[Y e^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]}\right)^2 \\ &=\; \mathbb{E}_{Q_\lambda}[Y^2] - \big(\mathbb{E}_{Q_\lambda}[Y]\big)^2 \;=\; \mathrm{Var}_{Q_\lambda}(Y),\end{aligned} \tag{4}$$

where the first step is the quotient rule applied to ψ' in Eq. (3), and the second step rewrites each ratio as an expectation under Q_λ and recognises the difference as a variance.

Step 3 (variance of a bounded variable). Under Q_λ the variable Y still takes values in $[a, b]$, since tilting only reweights the original law and does not enlarge the support. Writing $m := (a+b)/2$

for the midpoint, the variance is controlled by the second moment about m:

$$\begin{aligned} \mathrm{Var}_{Q_\lambda}(Y) &= \mathbb{E}_{Q_\lambda}\big[(Y-\mathbb{E}_{Q_\lambda}Y)^2\big] \le \mathbb{E}_{Q_\lambda}\big[(Y-m)^2\big] \\ &\le \left(\frac{b-a}{2}\right)^2 = \frac{(b-a)^2}{4}, \end{aligned} \tag{5}$$

where the first step is the definition of variance, the second uses that the mean minimises the mean-square deviation (so replacing $\mathbb{E}_{Q_\lambda}Y$ by the fixed m only increases the value), and the third uses the pointwise bound $|Y-m| \le (b-a)/2$, valid because $Y \in [a,b]$.

Step 4 (Taylor expansion). By Taylor's theorem with Lagrange remainder, there exists ξ between 0 and λ such that

$$\begin{aligned} \psi(\lambda) &= \psi(0) + \lambda\,\psi'(0) + \frac{\lambda^2}{2}\,\psi''(\xi) \\ &= \frac{\lambda^2}{2}\,\psi''(\xi) \le \frac{\lambda^2}{2}\cdot\frac{(b-a)^2}{4} = \frac{\lambda^2(b-a)^2}{8}, \end{aligned} \tag{6}$$

where the first step is the second-order Taylor expansion of ψ about 0, the second substitutes $\psi(0) = \psi'(0) = 0$ from Eq. (2) and Eq. (3), and the inequality combines Eq. (4) with the variance bound in Eq. (5). Exponentiating both sides yields Eq. (1). This completes the proof. □

2 Hoeffding's inequality

We now assemble the main result. The proof is the canonical Chernoff argument: exponentiate, apply Markov's inequality, factorise the moment generating function across the independent summands, bound each factor with Lemma 1.1, optimise the free parameter λ, and finally combine the two tails by a union bound.

Theorem 2.1 (Hoeffding's inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables with $X_i \in [a_i, b_i]$ almost surely for each i, where $a_i < b_i$, and set $S_n := \sum_{i=1}^n X_i$. Write $V := \sum_{i=1}^n (b_i - a_i)^2$. Then for every $t > 0$,*

$$\mathbb{P}\big[\,|S_n - \mathbb{E}[S_n]| \ge t\,\big] \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i-a_i)^2}\right). \tag{7}$$

Proof. Set $Y_i := X_i - \mathbb{E}[X_i]$, so that Y_i is centered with $Y_i \in [a_i', b_i']$ almost surely for $a_i' := a_i - \mathbb{E}[X_i]$ and $b_i' := b_i - \mathbb{E}[X_i]$; note $b_i' - a_i' = b_i - a_i$. The centered sum is $S_n - \mathbb{E}[S_n] = \sum_{i=1}^n Y_i$. The proof proceeds in three stages: a one-sided exponential tail bound, its optimisation over λ, and the two-sided combination.

Step 1 (Chernoff bound on the upper tail). For any $\lambda > 0$, the map $x \mapsto e^{\lambda x}$ is increasing and positive, so by Markov's inequality applied to $e^{\lambda(S_n - \mathbb{E}S_n)}$,

$$\begin{aligned} \mathbb{P}\big[\,S_n - \mathbb{E}[S_n] \ge t\,\big] &= \mathbb{P}\big[\,e^{\lambda(S_n-\mathbb{E}S_n)} \ge e^{\lambda t}\,\big] \\ &\le e^{-\lambda t}\,\mathbb{E}\Big[e^{\lambda\sum_{i=1}^n Y_i}\Big] = e^{-\lambda t}\prod_{i=1}^n \mathbb{E}\Big[e^{\lambda Y_i}\Big], \end{aligned} \tag{8}$$

where the first step exponentiates both sides of the event with the increasing bijection $x \mapsto e^{\lambda x}$, the second is Markov's inequality $\mathbb{P}[Z \ge s] \le \mathbb{E}[Z]/s$ applied to the nonnegative variable $Z =$

$e^{\lambda(S_n - \mathbb{E}S_n)}$, and the last step factorises the expectation of a product using the independence of $Y_1, \ldots, Y_n$ (a measurable function of independent variables).

Step 2 (apply Hoeffding's lemma and optimise λ). Each Y_i is centered and bounded in an interval of width $b_i - a_i$, so Lemma 1.1 bounds each factor by $\mathbb{E}[e^{\lambda Y_i}] \leq \exp(\lambda^2(b_i - a_i)^2/8)$. Substituting into Eq. (8) and writing $V = \sum_i (b_i - a_i)^2$,

$$\begin{aligned} \mathbb{P}\big[\, S_n - \mathbb{E}[S_n] \geq t \,\big] \;&\leq\; e^{-\lambda t} \prod_{i=1}^{n} \exp\!\left(\frac{\lambda^2 (b_i - a_i)^2}{8} \right) \\ &=\; \exp\!\left(-\lambda t + \frac{\lambda^2 V}{8} \right) \;=:\; \exp\big(g(\lambda)\big), \end{aligned} \tag{9}$$

where the first step applies Lemma 1.1 to each of the n independent factors, and the second collects the product of exponentials into a single exponent $g(\lambda) := -\lambda t + \lambda^2 V/8$. The bound holds for every $\lambda > 0$, so we minimise g. The first-order condition is

$$g'(\lambda) \;=\; -t + \frac{\lambda V}{4} \;=\; 0 \qquad \Longleftrightarrow \qquad \lambda^\star \;=\; \frac{4t}{V} \;>\; 0, \tag{10}$$

and $g''(\lambda) = V/4 > 0$ confirms $\lambda^\star$ is the global minimiser. Evaluating the exponent at $\lambda^\star$,

$$g(\lambda^\star) \;=\; -\frac{4t}{V}\, t + \frac{1}{8} \left(\frac{4t}{V} \right)^2 V \;=\; -\frac{4t^2}{V} + \frac{2t^2}{V} \;=\; -\frac{2t^2}{V}, \tag{11}$$

where the first equality substitutes $\lambda^\star = 4t/V$ into $g(\lambda) = -\lambda t + \lambda^2 V/8$, and the remaining equalities are arithmetic simplification. Combining Eq. (9) and Eq. (11) gives the one-sided bound

$$\mathbb{P}\big[\, S_n - \mathbb{E}[S_n] \geq t \,\big] \;\leq\; \exp\!\left(-\frac{2t^2}{V} \right). \tag{12}$$

Step 3 (two-sided bound by a union bound). Applying the one-sided bound Eq. (12) to the variables $-X_i \in [-b_i, -a_i]$, whose interval widths are again $b_i - a_i$ and whose centered sum is $-(S_n - \mathbb{E}S_n)$, yields the matching lower-tail bound

$$\mathbb{P}\big[\, S_n - \mathbb{E}[S_n] \leq -t \,\big] \;=\; \mathbb{P}\big[\, (\mathbb{E}S_n - S_n) \geq t \,\big] \;\leq\; \exp\!\left(-\frac{2t^2}{V} \right). \tag{13}$$

The two-sided event decomposes as $\{|S_n - \mathbb{E}S_n| \geq t\} = \{S_n - \mathbb{E}S_n \geq t\} \cup \{S_n - \mathbb{E}S_n \leq -t\}$. By a union bound over these two events, together with Eq. (12) and Eq. (13),

$$\begin{aligned} \mathbb{P}\big[\, |S_n - \mathbb{E}[S_n]| \geq t \,\big] \;&\leq\; \mathbb{P}\big[\, S_n - \mathbb{E}S_n \geq t \,\big] + \mathbb{P}\big[\, S_n - \mathbb{E}S_n \leq -t \,\big] \\ &\leq\; 2 \exp\!\left(-\frac{2t^2}{V} \right), \end{aligned} \tag{14}$$

where the first step is the union bound applied to the two tail events, and the second substitutes the upper- and lower-tail bounds. Since $V = \sum_{i=1}^{n} (b_i - a_i)^2$, this is exactly Eq. (7), which completes the proof. $\square$